Challenge 1: The Hadwiger Conjecture

Definition 1 (Graph Minor)**.** Given simple graphs X and G, we say that X is a *minor* of G if there is a function assigning to each vertex x of X a nonempty vertex set $V_x \subseteq V(G)$ such that

1. the subgraph $G[V_x]$ induced by G on V_x is connected for each x;
2. for different vertices $x, y \in V(X)$, the sets V_x and V_y are disjoint;
3. for any two adjacent vertices $x, y \in V(X)$, there are adjacent vertices $a \in V_x$ and $b \in V_y$ of G.

Definition 2 (Hadwiger Number)**.** Given a graph G, the *Hadwiger number* of G is the largest integer n such that K_n is a minor of G; here K_n denotes the complete graph on n vertices. We denote the Hadwiger number by $h(G)$.

Definition 3 (Chromatic number)**.** Given a graph G and $n \in \mathbb{N}$, a *proper coloring* of G with n colours is a function $f : [n] \to V(G)$ so that adjacent vertices receive different f-values. The *chromatic number* of G is the minimum number n such that there is a proper coloring of G with n colours.

Conjecture 1.1 (Hadwiger's Conjecture)**.** Let $r \in \mathbb{N}$. For every finite simple graph G, we have:

$$r \leq \chi(G) \implies r \leq h(G)$$

While Conjecture **??** is not difficult for every small values of r, already for $r = 4$ it is equivalent to the 4-colour theorem. Robertson, Seymour and Thomas proved the above for $r = 5$.